Lecture 27 : Project Scheduling Using PERT/CPM

Introduction

In the intricate world of project management, successfully guiding a project from conception to completion requires meticulous planning and foresight. Projects, by their very nature, are composed of numerous interconnected tasks, each with its own requirements and dependencies. Understanding how these tasks relate to one another, and how their individual durations impact the overall project timeline, is paramount for effective execution. This lecture delves into the foundational techniques of Project Scheduling: PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method).

These methodologies, developed over the last century, provide robust frameworks for breaking down complex projects, visualizing task relationships, and estimating project durations. By employing PERT and CPM, project managers gain invaluable insights into potential bottlenecks, critical activities that cannot be delayed, and areas where flexibility exists. This understanding empowers them to optimize resource allocation, mitigate risks, and ultimately deliver projects on time and within scope, making these techniques indispensable tools in any project manager's arsenal.

Core Concepts

Term	Definition	Significance
`Work Breakdown Structure (WBS)`	A hierarchical decomposition of a project into smaller, more manageable tasks, used to identify all necessary activities.	Forms the basis for identifying individual tasks and their relationships, crucial for network diagram creation.
`Task Network`	A graphical representation of project tasks and their interdependencies, showing which tasks must complete before others can start.	Visualizes the project flow, enabling analysis of task sequences and potential parallel execution.

Term	Definition	Significance
`Activity-on-Node (AON)`	A type of task network diagram where nodes represent activities and arrows represent the precedence relationships between them.	Highly intuitive and widely adopted for its simplicity in representing project activities and their dependencies.
`Activity-on-Arrow (AOA)`	A type of task network diagram where arrows represent activities and nodes represent the start or end events of activities.	Historically used but generally more complex than AON, sometimes requiring dummy activities to model relationships.
`PERT (Program Evaluation and Review Technique)`	A project scheduling technique developed for development projects with high uncertainty, using statistical estimates for task durations.	Accounts for variability in task times, providing probabilistic estimates of project completion, crucial for novel projects.
`CPM (Critical Path Method)`	A project scheduling technique developed for routine projects with predictable task durations, using deterministic time estimates.	Identifies the longest sequence of tasks (critical path) and the minimum project duration, vital for predictable projects.
`Critical Path`	The longest sequence of activities in a project network, determining the minimum possible time required to complete the project.	Highlights activities that, if delayed, will directly delay the entire project, requiring close monitoring.
`Slack (Float Time)`	The amount of time an activity can be delayed without delaying the project's overall completion time.	Indicates flexibility in scheduling non-critical activities, allowing for resource redistribution and risk management.
`Optimistic Time`		

Term	Definition	Significance
	The shortest possible time an activity can take, assuming everything goes perfectly.	Provides a best-case scenario estimate for task duration in PERT analysis.
`Pessimistic Time`	The longest possible time an activity can take, assuming significant problems or delays occur.	Provides a worst-case scenario estimate for task duration in PERT analysis.
`Most Likely Time`	The most probable time an activity will take under normal circumstances.	Represents the project manager's best guess for task duration in PERT analysis.

Detailed Analysis

1. Task Network Diagrams: Visualizing Project Flow

Task networks are fundamental to project scheduling, offering a visual representation of all project activities and their interdependencies. Over time, two primary types have gained prominence: Activity-on-Node (AON) and Activity-on-Arrow (AOA).

- **Activity-on-Node (AON) Diagrams:**
- In AON diagrams, each node (typically a rectangle or circle) represents a specific activity within the project.
- The name of the activity and its estimated duration are typically labeled within or adjacent to the node.
- Arrows connecting the nodes indicate precedence relationships, meaning an activity at the tail of an arrow must be completed before the activity at the head of the arrow can begin.
- AON diagrams are highly intuitive and widely adopted due to their straightforward representation.
- To simplify computations and ensure a clear project start and end, it is recommended to use a single dummy start node and a single dummy end node, even if the project naturally has multiple initial or final activities. This consolidates multiple starting or ending points into a single, unified entry and exit for the network.

- **Activity-on-Arrow (AOA) Diagrams:**
- In AOA diagrams, activities are represented by the arrows (or edges) connecting nodes.
- The nodes themselves represent events, signifying the start or completion of one or more activities.
- Activity names and durations are labeled on the arrows.
- AOA diagrams can be more complex to draw and interpret, often requiring the introduction of "dummy activities" (represented by dashed arrows with zero duration) to correctly model complex precedence relationships without implying actual work.
- Due to their increased complexity compared to AON, AOA diagrams are less commonly used in modern project management software and practices.
- **Drawing an Activity-on-Node Task Network Example:**
  Let's consider a set of activities with their predecessors:
- A: No predecessor
- B: Predecessor A
- C: Predecessor A
- D: Predecessor B
- E: Predecessor B
- F: Predecessor C
- G: Predecessor D
- H: Predecessors E and F

The process to draw this AON diagram would be:
* Start with activity A, as it has no predecessors.
* Draw B and C, with arrows originating from A, indicating A must complete before B and C can start.
* Draw D and E, with arrows originating from B.
* Draw F, with an arrow originating from C.
* Draw G, with an arrow originating from D.
* Draw H, with arrows originating from both E and F, signifying both must complete before H can start.
* Finally, connect any activities that have no successors to a single dummy end node to represent the project's completion.

2. PERT and CPM: Historical Context and Evolution

PERT and CPM emerged in the 1950s as distinct yet related project scheduling techniques, both building upon earlier tools like the Gantt chart.

- **Critical Path Method (CPM):**
- Developed by DuPont for managing maintenance projects in chemical plants.
- Characterized by `deterministic` task durations, meaning the time required for each activity is known with relative certainty. This is typical for routine, well-understood tasks like maintenance.
- Initially focused on identifying the critical path to determine the minimum project duration.
- Activities are often represented as rectangles or circles in network diagrams.
- **Program Evaluation and Review Technique (PERT):**
- Developed by the US Navy for the Polaris missile development project.
- Characterized by `statistical` or `probabilistic` task durations, acknowledging the inherent uncertainties in novel, research-and-development projects.
- Uses three time estimates for each activity: `optimistic time`, `most likely time`, and `pessimistic time`. These estimates allow for statistical inference regarding project completion time.
- Historically, PERT often used activity-on-arrow diagrams, but has largely adopted activity-on-node diagrams over time.
- **Convergence and Modern Usage:**
- While initially separate, PERT and CPM have largely merged into a single, comprehensive technique over the decades.
- Both are graphical, network-based methods that represent precedence relationships and activity sequences.
- Both aim to estimate project duration and identify critical activities.
- The key distinction remains in how they handle task durations: CPM assumes deterministic times, while PERT incorporates uncertainty through statistical estimates. However, modern applications often combine these approaches, allowing for both deterministic and probabilistic analysis within the same framework.

3. The Significance of Critical Path and Slack

Understanding the critical path and activity slack is paramount for effective project management.

- **Critical Path:**
- The critical path is the longest sequence of activities in the project network.
- It dictates the minimum possible time required to complete the entire project.
- Any delay in an activity on the critical path will directly result in a delay of the project's overall completion.
- Project managers must pay extra attention to critical activities, ensuring they stay on schedule.
- **Slack (Float Time):**
- Slack refers to the amount of time a non-critical activity can be delayed without affecting the project's overall completion date.
- Activities not on the critical path have some amount of slack.
- Activities with zero slack are by definition critical activities.
- Slack provides project managers with flexibility. They can strategically reallocate resources (e.g., moving manpower from a task with high slack to a critical task) to ensure critical activities remain on track, without jeopardizing the project schedule. This allows for efficient resource utilization and proactive risk management.

Key Takeaways

- [ ] Project scheduling techniques like PERT and CPM are essential for breaking down complex projects and managing task interdependencies.
- [ ] Activity-on-Node (AON) diagrams are the preferred method for representing task networks due to their intuitive nature and simplicity.
- [ ] A single dummy start and end node should be used in task networks to simplify computations, even with multiple initial or final activities.
- [ ] PERT and CPM, though historically distinct, have largely merged, with the main difference being their approach to task duration estimation (statistical for PERT, deterministic for CPM).
- [ ] PERT uses optimistic, most likely, and pessimistic time estimates to account for uncertainty in task durations, especially in development projects.

- [ ] The Critical Path is the sequence of activities that determines the shortest possible project duration; any delay on this path delays the entire project.
- [ ] Slack (or float time) indicates how much an activity can be delayed without impacting the project's overall completion, offering flexibility for resource management.
- [ ] Project managers use critical path and slack information to prioritize tasks, allocate resources effectively, and proactively manage potential delays.

Conclusion

The methodologies of PERT and CPM provide project managers with powerful analytical tools to navigate the complexities of project execution. By meticulously mapping out task networks, identifying critical paths, and understanding the inherent flexibility offered by slack, managers can gain unprecedented control over their projects. These techniques move beyond simple task lists, offering a dynamic framework for forecasting project completion, identifying potential bottlenecks, and making informed decisions about resource allocation.

The ability to distinguish between critical and non-critical activities, and to quantify the permissible delay for the latter, is a cornerstone of proactive project management. This strategic insight allows for the efficient deployment of resources, ensuring that the project stays on schedule even when unforeseen challenges arise. As we move forward, the application of these principles will be further enhanced by tools like the Gantt chart, which will be explored in subsequent lectures to demonstrate how these foundational scheduling concepts translate into real-time project monitoring and control.